

Homework — 1.5.2 Moral and Ethical Issues

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.5.2 lesson.

Part A — This week's topic (~14 marks)

1. A hospital introduces an AI system that reads scans and flags likely tumours for a radiologist to check.

(a) State **one** benefit of using automated decision making here. **[1]** (AO2)

(b) Explain **one** ethical concern with relying on this system, referring to a named issue from 1.5.2. **[1]** (AO2)

2. A facial-recognition system used by a police force is found to misidentify people from some ethnic groups far more often than others.

(a) Explain **why** an AI system can end up biased in this way. **[2]** (AO2)

(b) Explain why the "black box" nature of such a system makes the problem harder to deal with. **[1]** (AO1/AO2)

3. A tech company moves its software-development team from the UK to a country with lower wages (offshoring). State **one** benefit to the company and **one** drawback that this could cause. **[2]** (AO2)

A series of horizontal lines for handwriting practice. The first five lines are solid blue, and the last five lines are dashed blue.

7. (1.3.3) State the purpose of the **TCP** layer and the purpose of the **IP** layer in the TCP/IP stack. [2]
(AO1)

8. (1.1.2) State **one** reason a **multi-core** processor can complete a workload faster than a single-core processor of the same clock speed. [2] (AO1)
